

ABHIJITH SIVAPRASADAN

Thermal & Energy-Systems Modelling Engineer | CFD, Dynamic/Digital-Twin Simulation & Energy-Systems Optimisation (Python/MILP)

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PROFILE

Modelling-focused engineer with an M.Sc. in Sustainable Energy Engineering from KTH, building and verifying models across CFD, dynamic thermal/process simulation and energy-systems optimisation. My master thesis at Siemens Energy's Fluid Dynamics Laboratory built and verified compressible-flow and conjugate heat-transfer models of high-temperature industrial hardware in ANSYS Fluent. Outside coursework, I build and verify independent dynamic and optimisation models: a Modern Fortran gas-turbine/combined-cycle/grid-operation simulator with automated regression and scenario testing (ThermoTwin-F); Pyomo/HiGHS mixed-integer energy-systems models covering offshore wind-hydrogen-heat coupling (PyNEXUS) and exploratory grid-flexibility screening (PyPSA-NL); and Modelica/FMI co-simulated digital-twin demonstrators cross-checked against independent Python implementations (OpenSteamOpt, TES Discharge Screen). Independent cross-checking, through an analytical calculation, a second implementation, automated tests or measured data, is a recurring part of my modelling workflow, on a foundation of production software-engineering discipline from backend development work.

EDUCATION

M.Sc. Sustainable Energy Engineering Aug 2023 - 2026
KTH Royal Institute of Technology, Stockholm — 123 ECTS (120 required + 3 elective) complete per official KTH confirmation letter (24 Sep 2026); degree certificate/conferral in progress

B.Tech Mechanical Engineering 2017 - 2021
College of Engineering Perumon, APJ Abdul Kalam Technological University, India

PROFESSIONAL EXPERIENCE

Master Thesis Student, Fluid Dynamics Laboratory Apr - Nov 2025
Siemens Energy AB, Finspång

- Built compressible CFD and conjugate heat-transfer models in ANSYS Fluent (SST $k-\omega$) of high-temperature industrial hardware, verifying spatial-discretisation sensitivity with a three-level mesh-independence study; an analytical heat-loss consistency check agreed with Fluent within 0.1% across six CHT cases.
- When an internal heater failure made the planned experimental campaign impossible, conducted the root-cause analysis with the industrial supervisor and delivered a re-scoped 15-run numerical study within the original schedule.

Energy Efficiency Intern Dec 2024 - Jun 2025
Alleima AB (remote)

- Developed a comparative industrial energy-audit methodology across six countries (KPI/EnPI baselines, load-driver normalisation, ISO 50001/EED-aligned framework) — a modelling and data-normalisation exercise in its own right.

Engineer (backend software) Sep 2021 - Jun 2023
QBurst Technologies, Thiruvananthapuram, India

- Developed production backend services (Go, later TypeScript/NestJS) with automated tests; the reproducibility and verification discipline now underpins every model-vs-measurement and model-vs-model check I run.

MODELLING & ENERGY-SYSTEMS PROJECTS

ThermoTwin-F: gas-turbine, combined-cycle and grid-operation simulator Modern Fortran (Fortran 2008), independent project

- Built an inspectable simulator connecting component thermodynamics to plant operation: design-point performance, degradation/uncertainty studies, synthetic-fault inference, off-design/part-load behaviour and combined-cycle/HRSG modelling, plus grid-dispatch and frequency-response modules.
- Verified with an automated test suite (physics self-tests, workflow tests, CLI regression checks) and a reproducible analysis pipeline recording git revision, hashes and RNG seed; explicitly framed per its own documentation as a research/educational model, not calibrated or validated against a commercial engine or plant.

PyNEXUS: electricity-hydrogen-heat optimisation model Python, Pyomo/HiGHS, independent project

- Built a coupled offshore-wind, PEM-electrolyser and hydrogen-pipeline MILP with optional storage, grid and heat-recovery coupling, with a 168-hour reference configuration and a recorded 8,760-hour annual run using real ERA5 wind data, plus independent dispatch verification and reconciliation checks.

PyPSA-NL: exploratory grid-flexibility model Python, PyPSA/Linopy/HiGHS, independent project

- Built a Netherlands-inspired grid-flexibility screening model covering regional buses/transmission, renewables/backup/BESS siting and sizing sweeps, flexible-connection logic and an N-1 screening proxy; framed as exploratory, not a validated TSO-grade planning tool.

OpenSteamOpt: steam-and-power optimisation and Modelica digital-twin demonstrator Python, Pyomo/HiGHS, Modelica/FMI

- Built a two-boiler/two-header steam-and-power plant demonstrator coupling MILP dispatch optimisation with a Modelica dynamic plant model, verified via FMI 2.0 co-simulation (OpenModelica/FMPy) and cross-checked against an independent Python shadow-twin model.

TES Discharge Screen: thermal-storage technology screening framework

Python, Pyomo/HiGHS,
Modelica/FMI

- Built a packed-bed dynamic thermal model with molten-salt and PCM sub-models; cross-checked a Modelica FMU against an independent Python model over an 8-hour test (0.23% maximum temperature deviation).

Day-ahead machine-learning forecast of building heating demand

KTH (MJ2507), Grade A,
Python, group project

- Helped build a day-ahead heating-demand forecasting workflow for a real, approximately 3,600 m² KTH university building, covering multivariate data cleaning, Mahalanobis-distance outlier filtering and feature scaling, comparing a baseline model against shallow and deep neural networks on held-out building-operation data.

Numerical methods for heat-transfer and transport problems

KTH (MJ2515), Grade A, Python

- Implemented and compared numerical methods for heat-transfer and transport problems, including 1-D/2-D finite-difference conduction, transient time integration (Euler/RK4) and advection schemes; analysed discretisation error, residual convergence and stability against analytic/reference solutions.

Independent software projects above use AI-assisted implementation; I define the problem formulation, model architecture, validation criteria and interpretation of results.

TECHNICAL SKILLS

CFD & heat transfer	ANSYS Fluent (compressible flow, CHT, SST k- ω); mesh-independence methodology
Dynamic & digital-twin modelling	Modern Fortran, Modelica/OpenModelica, FMI 2.0 co-simulation; model-vs-model and model-vs-measurement verification
Energy-systems optimisation	Pyomo/HiGHS and PyPSA/Linopy MILP/LP formulation, dispatch and capacity-sizing problems
Data-driven modelling	Python (pandas, NumPy, scikit-learn), multivariate data cleaning/outlier detection, shallow/deep neural networks
Programming	Python, MATLAB, Modern Fortran, Go, TypeScript; Git, reproducible research workflows, automated testing

ADDITIONAL INFORMATION

Conference: Sponsored student delegate, Converging Energy Technologies (CET2026), Oskarshamn, Sweden, 28 Sep - 1 Oct 2026
Languages: English (professional working proficiency), Malayalam (native), Swedish (actively studying)